

Appendix 4: FKB327-004 protocol outline

Clinical Study Protocol

A Phase I, Randomized, Single-Blind, Single-Dose Study to
Compare Pharmacokinetic Characteristics and Safety of
FKB327 with those of Humira® in Japanese Healthy Subjects

Protocol Number	FKB327-004
Phase of development	Phase I
Objectives	The objective of the study is to compare pharmacokinetics (PK) and the safety of FKB327 and Humira® after single doses, by subcutaneous (SC) injection in Japanese healthy male subjects. Immunogenicity and local tolerability of FKB327 will also be investigated.
Study population	Japanese healthy male subjects
Number of subjects	130 subjects (FKB327: 65 subjects, Humira: 65 subjects) If PK parameters are not within the biosimilarity acceptance criterion because of an insufficient number of subjects, one add-on subject study can be performed using not less than half the number of subjects with the same methodology as the initial study in reference to the “Guideline for Bioequivalence Studies of Generic Products”.
Study drugs	<u>Test drug: Test formulation</u> Identification Code: FKB327 Dosage form: Injection (Pre-filled syringe) Specification/ content: 40 mg syringe 0.8 mL; each syringe contains 40 mg FKB327 <u>Reference drug: Reference formulation</u> Code: Humira® Product name: Humira® (US-licensed) International Nonproprietary Name (INN): Adalimumab Dosage form: Injection (Pre-filled syringe) Specification/content: 40 mg syringe 0.8 mL; each syringe contains 40 mg Adalimumab
Study design and methods	This is a phase I, randomized, active control, single-blind, parallel-group, clinical pharmacology study in Japanese healthy male subjects. The principal investigator and sub-investigators will perform a screening examination between Days -28 and -2 for the subject whose consents are obtained in order to confirm subject eligibility before dosing. Eligible subjects will be admitted to the trial site one day before dosing (Day -1) and undergo a pre-dose examination. The Investigators will stratify the eligible subjects by weight (< 65 kg and ≥ 65 kg) before dosing and will randomly assign subjects to the treatment group (65 subjects each in FKB327 and Humira group). The Investigators will administer a 40 mg single dose by SC injection in each subject according to the study drug allocation. A subject will discharge when he completes specified examinations and assessments on Day 9, and accordingly, if clinical abnormalities are not observed that require hospitalisation for follow-up and provided that the Investigators permit the discharge. Thereafter, the subject will revisit the trial site on specified days between Day 16 and 65 to take prescribed examination and assessments.

Major examinations and assessments	Subject background		Sex, age, birth date, medical history, prior treatment/therapy, concomitant drug/therapy confirmation of inclusion and exclusion criteria, and drinking and smoking habits.
	Physical findings		Subjective symptoms, physical examination, height, body weight, and body mass index (BMI).
	Tuberculosis test		Tuberculosis testing (T-SPOT)
	Vital signs		Blood pressure, pulse rate, respiration rate, and body temperature.
	12-lead ECG		12-lead ECG
	Laboratory tests	Hematology	Hemoglobin, RBC, MCV, MCH, MCHC, hematocrit, WBC, differential WBC (neutrophils, lymphocytes, monocytes, eosinophils, and basophils) and PLT.
		Biochemistry	BUN, creatinine, uric acid, total bilirubin, total protein, albumin, differential protein (albumin, α 1-globulin, α 2-globulin, β -globulin, γ -globulin), ALP, AST, ALT, GGT, CRP, fasting blood glucose, phosphorus, total cholesterol, triglyceride, potassium, sodium, calcium, and chlorine.
		Urinalysis (spot urine)	Protein, occult blood, ketone body, glucose, bilirubin, urobilinogen, specific weight, nitrite, PH, and leukocyte esterase.
	Drugs of abuse test		Phencyclidine and derivatives, benzodiazepine and derivatives, cocaine-derived narcotics, psychostimulants, morphine-derived narcotics, cannabis, barbiturate and derivatives, and tricyclic antidepressants.
	Infectious disease testing		Tests for HBsAg, anti-HCV antibody, and HIV antigen/antibody and qualitative tests for syphilis
	Anti-drug antibody		Anti-drug antibody
	Local tolerability		Injection site reaction and pain at the injection site
Endpoint	<u>Pharmacokinetics (PK):</u> 1) Primary endpoints C_{max} , AUC_{0-360h} and AUC_{0-t} 2) Secondary endpoints $AUC_{0-\infty}$, t_{max} , and $t_{1/2}$ <u>Safety:</u> • Adverse events (AEs) • Laboratory tests • Vital signs • 12-lead ECG • Anti-drug antibody (ADA) • Local tolerability; – Injection site reaction – Visual analogue scale (VAS) for pain at injection site		
Statistical analysis	<u>PK Analysis:</u> Summary statistics will be calculated for PK parameters using serum concentrations samples treated by either test and or reference drugs. For biosimilarity evaluation, primary PK parameters (C_{max} , AUC_{0-360h} , and AUC_{0-t}) will be log-transformed and analyzed. Biosimilarity will be concluded if 90% CI of mean difference in each log-transformed PK parameters of test and reference drugs is within the range of $[\log(0.80)-\log(1.25)]$.		

	<u>Safety Analysis:</u> 1) AE All AEs will be listed in a table. Number of subjects and frequency of treatment-emergent adverse events (TEAEs) will be counted by treatment group for preferred term and system organ class of MedDRA. In addition, adverse drug reactions will be counted similarly. 2) Laboratory safety variables The data for hematology, biochemistry, and urinalysis will be summarized in a table. For continuous variables, basic statistics will be calculated by treatment group at each time point. For nominal and ordinal scale variables, a shift tables will be created by treatment group for each measurement point of pre and post dosing. 3) Vital sign The results for vital sign will be summarized in a table. The vital sign data at each measurement point will be summarized and descriptive statistics will be generated by treatment group. 4) 12-lead ECG The results of 12-lead ECGs will be summarized in a table. The ECG measurements will be summarized in degrees and percentage at each time point of measurement by treatment group. 5) Anti-drug antibody (ADA) The results for ADA test will be summarized in a table. ADA expression rate and titer will be listed at each measurement point. 6) Injection site reaction The score and the size of injection site reaction (mm) will be summarized in a table. The scores at each measurement point will be summarized in a table by frequency, percentage by treatment group. In addition, injection site reaction size (mm) will be calculated at each point by treatment group and summarized in a table at each point on an ordinal scale by degree, percentage by treatment group. 7) Injection site pain using VAS Injection site pain scores using VAS will be listed in a table. Descriptive statistics will be calculated by treatment group at each measurement point.
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TIME AND EVENTS SCHEDULE

Study day (Day)		-28 to -2	-1	1					2		3	4	5	6	7	8	9	16	23	30	37	44	51	65 or early termination	
Theoretical time (hour)	Consent	Screening		Pre-dose	0	0.5	1	4	12	24	36	48	72	96	120	144	168	192	360	528	696	864	1032	1200	1536
Informed consent	●																								
Medical history/ demographics		●																							
Physical examination, Vital signs, ECG, Laboratory testing are assessed at the schedule defined by protocol		←																							→
Dosing					●																				
Local injection site reaction					● ¹⁾				●	●		●	●	●											
VAS for pain (injection site)					● ¹⁾	●	●		●	●															
Blood sampling for ADAs				●															●		●				●
Blood sampling for PK				●				●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Adverse events					←																				→

1) Immediately after dosing